

Displays

LCD Displays: The very first speed bump in the road to miniaturisation is a device's display. The fact is that today's mobile manufacturing technologies are easily capable of cramming currently offered features into devices that are much, much smaller than the devices of today. The problem begins with the popular LCD display technologies that are incorporated into almost all devices today.

It's not that LCD technologies are bad, or cannot be made smaller—the problem lies with human sight. You can only shrink displays to a certain size, after which they are no longer comfortably viewable by the naked eye. Since mobile phones are fast becoming all-in-one solutions, which incorporate audio-visual content such as stored movie clips, streaming video, MP3 playback, video cameras and more—it's only natural that a decent sized screen is needed.

The near future might just see the end of LCDs, in their current form, in mobile devices. However, in order to do away with LCD screens we first need to develop viable alternatives.

Paper Displays: We have looked at paper displays in earlier articles in *Digit*, and only recently have these displays become viable for use in mobile devices. Paper displays can be folded, bent and rolled, and this gives them an immediate advantage over LCD displays. However, as of yet, refresh rates are a problem, and paper displays are still more suited for static content such as advertisements or e-books. Colour video is not really possible as of yet, but researchers feel that it's only a matter of time before this becomes a reality.

Many sceptics wrote off research in paper displays as a waste of time, and were also not convinced that colour paper displays would ever be made. Today, not only does the technology exist, but products and working prototypes are already available. It's true that these displays cannot replace the LCDs on high-end mobiles yet. Nevertheless, they could be used in entry-level devices.